ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician Basic Maintainer (00379)

USQ Roundsperson KIN (ALMK)

MITE Course IDs:
On-board Phase – 121181

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RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commandant of the Curriculum Control Authority CCA (NFS(P)). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J “Course History”.

Revision	Approval Date	Authority
1.0	10 May 2024	CNTG

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FOREWORD

1. This training document is developed under the authority of the Commander Naval Training and Personnel Group (CNPTG) and in accordance with the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-002 Non-Commissioned Member General Specification dated 20 NOV 20;

   c. A-PD-055-002/PP-0021 MAR TECH – MOS 00379 Occupational Specification dated 25 MAR 24; and

   d. NAVORD 4500-0 - Structured Delegation, and NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA, who will forward to NPTG for approval, as appropriate.

5. The designated TE(s) for this training is indicated below:

TE	Phase/Module	MITE Course ID	NQual
NFS(P)	All	121181	ALMK
NFS(A)	All	121181	ALMK

CHAPTER 1: **TRAINING MANAGEMENT DETAILS**

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED
BY
COMMANDER NAVAL TRAINING GROUP (CNTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Roundsperson on a KIN class vessel. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

PREREQUISITES

2. Prerequisites. To be loaded on this course OJPR trainees must have the following prerequisites:

   a. BMQ
   b. RQ-S3 Marine Technician MITE ID: 122592 NQual ALJS

3. Instructor Qualifications. To conduct this training instructors must hold a minimum of one of the following qualifications:

   a. USQ – ENG Watchkeeper KIN Class ALVB; or
   b. MS Cert 2K (legacy qual).

QUALIFICATION

4. Successful completion of this QSP-OJPR constitutes the primary means of attaining the Roundsperson KIN class NQual ALMK.

SUCCESSFUL COMPLETION REPORTING

5. Upon successfully completing this NQUAL Certificates and NPTORD 4500-51 - Course Reports and Certification.

EVALUATION

6. Program evaluation shall be conducted IAW the guidance provided in A-P9050-000 CFITES Vol. 11 - Evaluation and Validation Techniques.

VALIDATION

7. This training will be validated every five years CFITES Vol. 8 - Validation of Instructional Programs and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

8. PLARs shall be conducted in accordance with NPTORD 4500-27 – PLAR, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

9. The Kingston USQ Roundsperson manual should be provided at the time the member receives the OJPR.

CHAPTER 2: **PERFORMANCE OBJECTIVES**

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL TRAINING GROUP (CNTG)

PO 001 PERFORM AUXILIARY SYSTEM ROUNDS

1. Performance. Perform Auxiliary System Rounds.

2. Conditions

   a. Given:

      (1) Kingston class ship;
      (2) PPE;
      (3) Drawings;
      (4) Log books;
      (5) Supervision;
      (6) References; and
      (7) Tools and test equipment.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Roundsperson on a KIN class vessel shall Perform Auxiliary System Rounds IAW refs by:

   a. Inspecting liquid handling systems by:

      (1) Monitoring a liquid level through CAMs; and
      (2) Sounding Tubes and Dip tape.

   b. Inspecting domestic services by:

      (1) Inspecting visually, fresh water systems to include:

         i. Flash up fresh water system, locally;
         ii. inspecting visually, potable water purification (bromine) injection system; and
         iii. Flash up reverses osmosis desalination (ROD) unit.

      (2) Configure black and grey water collection system to include;

         i. Configure black and grey water systems for collection;
         ii. Configure black and grey water systems for discharge; and
         iii. Treatment of black and grey water.

(3) Operating fluid piping systems to include:

i. Freshwater;
ii. Clorifiers;
iii. Black water; and
iv. Grey water.

c. Inspecting compressed air systems by:

(1) Configuring compressed air system locally;
(2) Flash up portable air compressor.

d. Inspecting refrigeration and HVAC system to include:

(1) Inspecting main refrigeration system;
(2) Configuring HVAC; and
(3) Flash up main refrigeration system locally.

4. Time Required.

a. PO – 1575 min; and

b. PC – OJPR.

5. Instructional Methods.

a. Demonstration – performance;

b. Self-study; and

c. OJT.

6. Training Aids. Nil.

7. Training Limitations. Ship availability.

8. Test Details.

a. PO 001 will be assessed by PC 001 IAW Annex K.

9. References. Specified in applicable EOs.

10. Remarks. Nil.

PO 002 PERFORM ELECTRICAL SYSTEM ROUNDS

1. Performance. Perform Electrical System Rounds.

2. Conditions.

   a. Given:

      (1) Kingston class ship;
      (2) PPE;
      (3) Drawings;
      (4) Log books;
      (5) Supervision;
      (6) References; and
      (7) Tools and test equipment.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Roundsperson on a KIN class vessel shall Perform Electrical System Rounds IAW refs by:

   a. Inspecting PG&D by:

      (1) Operate PG&D system to include:

         i. operating the power generation system remotely; and
         ii. operating the 450V distribution system.

      (2) Operating electrical switchboard locally and remotely to include:

         i. operating electrical switchboard locally;
         ii. paralleling electrical loads locally (shore power and ship);
         iii. paralleling electrical loads remotely; and
         iv. operating switchboard breakers remotely.

      (3) Testing UPS to include:

         i. Inspecting uninterruptable power supplies wiring harness.

      (4) Operating motor alternator set; and

(5) Testing Man Overboard Alarm.

b. Inspecting electrical systems by:

(1) Inspecting whistle assemblies to include:

i. testing ship's whistle; and
ii. testing whistle speed controllers.

(2) Inspecting whistle at will switches to include:

i. testing whistle at all switches.

(3) Operating central alarm monitoring system (CAMS) to include:

i. testing H2S and IFDMS alarm relays ;
ii. testing man overboard system;
iii. testing man locked in fridge alarm system; and
iv. testing Para Sense.

(4) Operating Navigational lights locally.

c. Inspecting cathodic & degaussing system by:

(1) Operating cathodic protection system to include:

i. conducting hull potential test; and
ii. switching unit between modes.

(2) Testing degaussing rectifier controllers.

d. Inspecting motor and controls by:

(1) Inspecting, electric motor (AUT1690);

(2) Flashing up motor controllers locally and remotely.

4. <u>Time Required.</u>

a. PO – 1450 min; and

b. PC – OJPR.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Self-study; and

   c. OJT.

6. Training Aids. Nil.

7. Training Limitations. Ship availability.

8. Test Details.

   a. PO 002 will be assessed by PC 002 IAW Annex K.

9. References. Specified in applicable EOs.

10. Remarks. Nil.

PO 003 PERFORM DC SYSTEM ROUNDS

1. Performance. Perform DC System Rounds.

2. Conditions.

   a. Given:

      (1) Kingston class ship;
      (2) PPE;
      (3) Drawings;
      (4) Log books;
      (5) Supervision;
      (6) References; and
      (7) Tools and test equipment.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Roundsperson on a KIN class vessel shall Perform DC System Rounds IAW refs by:

   a. Inspecting DC systems by:

      (1) Configuring fitted firefighting controls;
      (2) Inspecting visually, smoke control and evacuation system components to include:

         i. test smoke detector and pull stations;
         ii. operate Main Machinery Space Cameras; and
         iii. configure RAMFAN and trunking.

      (3) Configuring bilge eductor and bilge and ballast pump locally.

   b. Inspecting firemain system by:

      (1) Flashing up seawater service system (firemain) locally.

4. Time Required.

a. PO – 575 min; and

b. PC – OJPR.

5. Instructional Methods.

a. Demonstration – performance;

b. Self-study; and

c. OJT.

6. Training Aids. Nil.

7. Training Limitations. Ship availability.

8. Test Details.

a. PO 003 will be assessed by PC 003 IAW Annex K.

9. References. Specified in applicable EOs.

10. Remarks. Nil.

PO 004 PERFORM PROPULSION SYSTEM ROUNDS

1. Performance. Perform Propulsion System Rounds

2. Conditions.

   a. Given:

      (1) Kingston class ship;
      (2) PPE;
      (3) Drawings;
      (4) Log books;
      (5) Supervision;
      (6) References; and
      (7) Tools and test equipment.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Roundsperson on a KIN class vessel shall Perform Propulsion System Rounds IAW refs by:

   a. Inspecting fuel & oil systems by:

      (1) Flash up fuel oil system locally to include:

         i. configure centrifuge; and
         ii. configure coalescing filter unit.

      (2) Transfer lube oil with lube oil filling & transfer system locally; and
      (3) Configure bilge pumping system (dirty oil pump).

   b. Inspecting shaft & gearing by:

      (1) Verify steering operations in follow up and non-follow up mode;
      (2) Flash up z-drive hydraulic oil (steering) system locally (XT0100).
      (3) Flash up Z-drive lube oil system locally (ABT0240);
      (4) Operating main shaft line brake in local (ABT0180); and

      (5) Configure main seawater circulation system locally (DT0190).

c. Inspecting combustion engines by:

(1) Flashing up Auxiliary Alternator;
(2) Flashing up Main Alternator;
(3) Flashing up Emergency Alternator; and
(4) Inspecting dirty oil system.

4. Time Required.

a. PO – 1050 min; and

b. PC – OJPR.

5. Instructional Methods.

a. Demonstration – performance;

b. Self-study; and

c. OJT.

6. Training Aids. Nil.

7. Training Limitations. Ship availability.

8. Test Details.

a. PO 004 will be assessed by PC 004 IAW Annex K.

9. References. Specified in applicable EOs.

10. Remarks. Nil.

PO 005 RESPOND TO EMERGENCY CONDITIONS

1. Performance. Respond to Emergency Conditions

2. Conditions.

   a. Given:

      (1) Kingston class ship;
      (2) PPE;
      (3) Drawings;
      (4) Log books;
      (5) Supervision;
      (6) References; and
      (7) Tools and test equipment.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Roundsperson on a KIN class vessel shall Respond to Emergency Conditions IAW refs by:

   a. Conducting emergency engineering drills;

   b. Responding to Action stations; and

   c. Responding to Emergency stations.

4. Time Required.

   a. PO – OJPR; and

   b. PC – OJPR.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Self-study; and

   c. OJT.

6. Training Aids. Nil.

7. Training Limitations. Ship availability.

8. Test Details.

   a. PO 005 will be assessed by PC 005 IAW Annex K.

9. References. Specified in applicable EOs.

10. Remarks. Nil.

CHAPTER 3: **ASSESSMENT OF TRAINEE**

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT OF THE CCA (NFS (P))

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of a practical OJPR and the trainee must obtain a pass in all POs to successfully achieve the Kingston Class Roundsperson NQual ALMK.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee and the ships training department to recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the ships training department and trainees. Each EO must have at least one EC.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation.

5. Enabling Checks are the responsibility of the Instructional Ships Staff (Qualified Ships Staff).

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional and standards staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programmes and TE SOs or SOPs.

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

    a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the TRB or PRB (if required); and

d. Information to assist in completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment SOs or SOPs.

TRAINING REVIEW BOARD

9. TRBs shall be convened and conducted IAW with NPTORD 4500-54 - Training Review Board Process.

COURSE REPORTS

10. Course reports (CF 377) shall be prepared and distributed IAW CFAO 26-12, 4500-51 - Course Reporting and Certificates.

11. Trainee will be graded pass or fail. IAW NPTORD 4500-51 - Course Reporting and Certificates, students will not be ranked.

12. Where necessary for Rank Qualifications, the narrative will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and/or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the training establishment.

CHAPTER 4: **LESSON SPECIFICATIONS**

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT OF THE CCA (NFS (P))

PO 001 Perform Auxiliary System Rounds

EO 001.01 Inspect Liquid Handling Systems

1. Performance. Inspect Liquid Handling Systems.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log books.

   b. Denied: Nil.

   c. Environmental: Ship alongside/at sea.

3. Standard. The trainee shall Inspect Liquid Handling Systems, IAW refs, by:

   a. Produce a liquid cargo report by monitoring a liquid level management system via computer alarm and monitoring system (CAMS) and sounding tubes (CT0270).

4. Time Required

   a. Completed during OJPR.

5. Instructional Methods:

   a. Demonstration – performance;

   b. Study assignment;

   c. Self-study;

   d. OJT.

6. Training Aids:

   a. CAMS;

   b. MCR administration computer; and

   c. Operational MCDV.

7. Training Limitations. Ship availability.

8. Task Numbers. CT0270.

9. Test Details.

   a. There is no formal EC for EO 001.01. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A11, A12, A13.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Deliver a Tour and Operate Tank Level Monitoring Systems					
1.1 Local Level monitoring: a. CAMS / Sound tubes / level indicators / LSU alarms and their locations to include: (1) Ballast; (2) Potable water; (3) Blackwater; (4) Greywater; (5) Void; (6) DFO; (7) Dirty oil; (8) Oily water; and Lube Oil storage;	D	AP	P	A11	Qualified SS to conduct ship tour pointing out all tank covers, sounding tubes and LSU alarm panels for tanks.
1.2 Remote level monitoring: a. CAMS	D	AP	P	A12	Qualified SS to demonstrate the use of CAMS to monitor LLM.

2.3 Describe purpose and produce a liquid cargo report.	D	AP	P	A13	

EO 001.02 Inspect Domestic Services

1. Performance. Inspect Domestic Services.

2. Conditions

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) cable log books.

   b. Denied: Nil.

   c. Environmental: Ship alongside or at sea.

3. Standard. The trainee shall Inspect Domestic Services, IAW refs, by:

   a. Inspecting visually, fresh water systems to include (JT0100);

      (1) Flash up fresh water system, locally (JT0260);
      (2) Inspecting visually, potable water purification (bromine) injection system (JT0130); and
      (3) Inspecting system operation of reverses osmosis desalination (ROD) unit (AOT0830).

   b. Configuring black and grey water collection system to include (JT0160);

      (1) Configure black and grey water systems for collection;
      (2) Configure black and grey water systems for discharge; and
      (3) Treatment of black and grey water.

   c. Operating fluid piping system (ATT0920) to include:

      (1) Freshwater;
      (2) Clorifiers;
      (3) Black water; and

(4) Grey water.

4. Time Required.

a. Completed during OJPR.

5. Instructional Methods.

a. Demonstration – performance;

b. Self-study; and

c. OJT.

6. Training Aids.

a. Operational MCDV.

7. Training Limitations. Ship Availability

8. Task Numbers. AOT0830, ATT0920, JT0100, JT0130, JT0160, JT0260

9. Test Details.

a. There is no formal EC for EO 001.02. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A14, A16, A17, A18

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP2 Deliver a Tour and Operate the Domestic Fresh Water Systems					

2.1 Freshwater Filling and Storage system to include: a. Upper deck filling valves; b. Upper deck riser isolation valves; and c. Tank filling and suction valves.	D	AP	P	A14	Qualified SS to demonstrate location and use of system.
2.2 Freshwater Distribution system to include: a. Pumps and local control panel; b. Local Pressure gauges; c. Normal System pressure; d. Procedure for changing over tanks; e. Pump supply & discharge valves; f. Procedure for priming and flashing pumps; g. Priming tank and valves; h. Supply and return isolation valves; i. Calorifiers; j. Relief valves; and k. Cold weather precautions.	D	AP	P	A14	Qualified SS to provide operating parameters and to demonstrate procedures.
2.3 Bromine Injection System to include: a. Location of equipment; b. Supply & return valves for tank 1 & 2; and c. Adjusting and testing bromine levels.	D	AP	P	TBP	Qualified SS to demonstrate operation and location of system.
3.4 Reverse Osmosis Desalination (ROD) a. Location of valves; b. Changing Cartridge filters; c. Alarm points; d. Trips; e. Conducting rounds; f. Flash-up procedure; and g. Shutdown procedure.	D	AP	P	A16	Qualified SS to demonstrate operation and location of system. Alarms and trips to be explained. Ensure to show log entries as required.
TP4 Deliver a Tour and Operate the Black and Grey Water Systems					

4.1 Location of system equipment to include: a. Local control panels; b. Eductor pumps; c. Discharge pumps d. Media tanks e. Wet well f. Circ tank g. Air blower h. Macerators i. Transformers j. Chlorinator k. Ultra violet sterilizer	D	AP	P	A17	Qualified SS to conduct a ship tour of all components. Pre sail checks and flash up of system will follow SOP's. (Lists procedures)
4.2 Location of system valves to include: a. Upper deck discharge b. Flushing connection c. Media tank isolation d. Collection e. Chlorine treatment system f. Ultra violet sterilization system	D	AP	P	A17	
4.3 Operation of the Blackwater System to include: a. Switching between modes of operation (collection, treatment, chlorine/UV) b. Setting up for upper deck discharge/ashore operation; and c. Logging operation changes.	D	AP	P	A17	Qualified SS to demonstrate configuration for different modes of operation. Ensure to show log entries as required.
4.4 Grey water components to include: a. Surge tank b. Grease trap c. Contact tank d. Chlorinator unit e. Storage tank (SW ballast tank #1) f. Discharge pump g. Quick closing valves	D	AP	P	A18	Qualified SS to conduct a ship tour of all components
4.5 Greywater System operation to include: a. Switching modes between collection and overboard discharge; b. Changing Chlorine pucks and cleaning the greywater pump strainer; and c. Logging the changes of mode.	D	AP	P	A18	

EO 001.03 Inspect Compressed Air System

1. Performance. Inspect Compressed Air System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log books.

   b. Denied: Nil.

   c. Environmental: ship alongside / at sea.

3. Standard. The trainee shall Inspect Compressed Air System, IAW refs, by:

   a. Configuring compressed air system locally (ET0170); and

   b. Flashing up portable air compressor locally (IT0560).

4. Time Required.

   a. Completed during OJPR.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Self-study; and

   c. OJT.

6. Training Aids.

a. Operational MCDV.

7. Training Limits. Ship Availability.

8. Task Numbers. ET0170, IT0560.

9. Test Details.

a. There is no formal EC for EO 001.03. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A19, A20, A21.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP3 Deliver a Tour and Operate the HP/LP Air System					
3.1 Locations of: a. Air compressors; b. Pressure vessels; Black water scouring supply; Air start Main D/As; c. Dampers; d. Shaft brakes; e. Thyristor pumps priming; f. Reducing stations g. Service outlets h. Bilge pump priming; i. Ship side valve's air supply; and j. Cross connect.	D	AP	P	A19	Qualified SS to conduct walkthrough of system, locating key components.
3.2 HP/LP Air System Operation: a. Flash-up of air compressors; b. Alarms and trips; c. Oil type; d. Lead / Lag set points; e. Changing over lead and lag compressors; f. Normal system pressures; and g. Blowing down of air system.	D	AP	P	A19, A20	Qualified SS to demo flash-up of compressor and rounds for the system.
TP4 Deliver a Tour and Operate the Breathing air compressors					
4.1 Bauer portable compressor operation	D	AP	P	A21	Qualified SS to walkthrough and show

					trainees operation, maintenance and logs.

EO 001.04 Inspect Refrigeration & HVAC

1. Performance. Inspect Refrigeration & HVAC.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision: and
      (7) Applicable log books.

   b. Denied: Nil.

   c. Environmental: Ship alongside or at sea.

3. Standard. The trainee shall Inspect Refrigeration & HVAC , IAW refs, by:

   a. Inspecting main refrigeration system (JT0120); and

   b. Configuring HVAC system, and

   c. Flashing up refrigeration system.

4. Time Required.

   a. System Walkthrough – 100 min; and

   b. System Operation – 75 min.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Self-study; and

   c. OJT.

6. Training Aids

   a. Operational MCDV.

7. Training Limitations.

8. Task Numbers.

9. Test Details.

   a. There is no formal EC for EO 001.04. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A20, A22, A23, A24, A25.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP2 Deliver a Tour and Operate the Main Refrigeration System					
2.1 Location of: a. Compressor b. Oil separator c. Condenser d. Drier e. Accumulator f. Regulating valve g. Evaporators h. Receiver i. Para Sense	D	AP	P	A20, A22	Qualified SS to conduct walkthrough on location of key components, demonstrate rounds, normal operation and leak testing on the main refrigeration system;
2.2 Main Refrigeration Operation : a. Starting and stopping refrigeration system b. Rounds on refrigeration system; and c. Trips and alarms					
2.3 Leak testing and ODS log					
TP3 Deliver a Tour and Operate the Heating Ventilation and Air Conditioning Systems (HVAC)					

3.1 HVAC Operation and Component Locations: a. MAC 1 & 2 Locations; b. Filter locations and types; c. Ventilations crash stops; d. Smoke evacuation dampers; e. Machinery space ventilation; and f. Local ventilation fan controllers	D	AP	P	A20, A23, A24, A25	Qualified SS to conduct walkthrough on location of key components and operation of HVAC.

PERFORMANCE CHECK – PO 001

Trainee will be provided feedback through PO 001. To achieve the intended assessment of PC 001, the trainee must achieve a qualifying signature in every signature block associated to PO 001 in the OJT package found in Annex E.

PO 002 Perform Electrical System Rounds

EO 002.01 Inspect Power Generation and Distribution

1. Performance. Inspect Power Generation and Distribution

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log books

   b. Denied: Nil.

   c. Environmental: Ship alongside or at sea.

3. Standard. The trainee shall Inspect PG&D, IAW refs, to include:

   a. Operating PG&D system (PT0270) to include:

      (1) Operating power generation system remotely (PT0290); and
      (2) Operating 440V distribution system (PT0120).

   b. Operating electrical switchboard remotely (PT0160) by:

      (1) Operating electrical switchboard locally (PT0170);
      (2) Paralleling electrical loads locally (AUT1950);
      (3) Paralleling electrical loads remotely (AUT1940); and
      (4) Operating switchboard breakers remotely (PT0160).

   c. Testing UPS by (AUT3900) by:

      (1) Inspecting uninterruptable power supplies wiring harness (AUT1630).

   d. Operating motor alternator (MA) to include:

(1) Testing switch unit auto (AUT3750); and
(2) Testing switch unit manual (AUT3770).

4. Time Required.

a. Completed during OJPR.

5. Instructional Methods.

a. Demonstration – performance;

b. Self-study; and

c. OJT.

6. Training Aids.

a. Operational MCDV.

7. Training Limits. Ship Availability

8. Task Numbers. AUT1630, AUT1940, AUT1950, AUT3750, AUT3770, AUT3900, PT0120, PT0160, PT0170, PT0270, PT0290.

9. Test Details

a. There is no formal EC for EO 002.1. The content within this EO will be assessed through the signature block found in Annex E.

10. References. A11, A20, A27, A28.

11. Remarks. Nil.

12. Teaching Points

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Deliver a shipboard tour of the PG&D system					

1.1 Deliver a shipboard tour to encompass: a. Diesel Alternators (Main, Aux, Emg); b. The MPPS, SS, and emergency switchboards; c. Thyristor Cabinets; d. Propulsion motors; e. Motor Alternator; f. Circuit Breakers (locations) and switchboard links; g. Shore connection; h. Fitted UPS locations; i. The anti-blackout system; j. CAMS and PGSCS console; k. Power generation and supervisory control system; and l. Local Scanning Units (LSU).	D	AP	P	A11, A20, A27, A28	The ships Electrician is to provide the tour of the switchboard and its components i.e. breakers.
TP2 Operate the PG&D system					
2.1 MPP and SS Switchboard operation: a. Receive power to a dead bus; b. Synchronize DA with DA; c. Synchronize DA and shore power (no check); d. Receive power to a dead bus; e. Synchronise MA & Aux DA; f. Synchronize Aux DA and shore power; and g. Synchronize MA and shore power.	D	AP	P		The ships Electrician is to provide oversight for this EO.
2.2 Anti-blackout system operation.	D	AP	P		The ships Electrician is to provide oversight for this EO.
2.3 SCR operation: Determine alarm condition and acknowledge; a. Troubleshoot alarm faults; and b. Operate in local control.	D	AP	P		The ships Electrician is to provide oversight for this EO.
2.4 Emergency Switchboard operation: a. Receive power to a dead bus; b. Synchronize Emergency DA and ship's service switchboard bus during split configuration; c. Reset/Restore switchboard and Emerg DA after Black-out sequence.	D	AP	P		The ships Electrician is to provide oversight for this EO.

2.5 Operate ground fault detection system on all power distribution system: a. 600Vac distribution; b. 450Vac distribution; c. 230Vac distribution; d. 115Vac distribution; and e. 24Vdc distribution.	D	AP	P		The ships Electrician is to provide oversight for this EO.
2.6 Isolate ground fault: a. Seek authorization for isolating power throughout the ship; b. Follow isolation procedure at both the switchboards and the power distribution panels; and c. Isolate circuit using tagout / lockout procedure.	D	AP	P		The ships Electrician is to provide oversight for this EO.
2.7 Isolation of electrical power in response to emergencies in: a. Home port; and b. Foreign port.	D	AP	P		The ships Electrician is to provide oversight for this EO.

EO 002.02 Inspect Electrical Systems

1. Performance. Inspect Electrical Systems.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log books.

   b. Denied: Nil.

   c. Environmental: Ship alongside or at sea.

3. Standard. The trainee shall Inspect Electrical Systems, IAW refs, by:

   a. Inspecting whistle assemblies (AUT1780) by:

      (1) Testing ship's whistle (AUT3700); and
      (2) Testing whistle speed controllers (AUT3950).

   b. Inspecting whistle at will switches (AUT1790) by:

      (1) Testing whistle at all switches (AUT3940).

   c. Operate central alarm monitoring system (ABT0100) by:

      (1) Testing alarm relay outputs (BCT0470);
      (2) Testing man overboard system (AOT1620); and
      (3) Testing man locked in fridge alarm system (AOT1610).

   d. Operating Navigational lights locally (PT0250).

4. Time Required.

a. Completed during OJPR.

5. Instructional Methods.

a. Demonstration – performance;

b. Self-study; and

c. OJT.

6. Training Aids.

a. Operational MCDV.

7. Training Limitations. Ship Availability.

8. Task Numbers. ABT0100, AOT1610, AOT1620, AUT1780, AUT1790, AUT3700, AUT3940, AUT3950, BCT0470, PT0250.

9. Test Details.

a. There is no formal EC for EO 002.02. The content within this EO will be assessed through the signature block found in Annex E.

10. References. A11, A29, A30.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Deliver a Tour and Operate the Whistle/Foghorn System					
1.1 Whistle Components to include: a. At-will switch location b. Whistle auto controller c. Feeder circuits d. Interconnection to navigational light system	D	AP	P	A29, A30	Combined homework assignment for all TPs

1.2 Operate fog horn/whistle to include: a. At-will switches Timing circuit b. IFDMS integration i.e. unmanned switch and lights c. Navlight interconnection	D	AP	P	A29, A30	
TP2 Operate the Shipboard Alarms					
2.1 Verify shipboard alarms by testing: a. Man over board, b. Man in locked fridge c. IFDMS (fire alarm)	D	AP	P	A11	

EO 002.03 Inspect Cathodics & Degaussing System

1. Performance. Inspect Cathodics & Degaussing System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log books.

   b. Denied: Nil.

   c. Environmental: Ship alongside or at sea.

3. Standard. The trainee shall Inspect Cathodics & Degaussing System, IAW refs, by:

   a. Operating cathodic protection system (LT0050) by:

      (1) Conducting hull potential test (AT0020); and
      (2) Inspecting shaft grounding system (AUT1440).

   b. Testing degaussing rectifier controllers (APT0470).

4. Time Required.

   a. Completed during OJPR.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Self-study; and
   c. OJT.

6. Training Aids.

   a. Operational MCDV.

7. Training Limits. Ship Availability.

8. Task Numbers. APT0470, AT0020, AUT1440, LT0050.

9. Test Details.

   a. There is no formal EC for EO 002.03. The content within this EO will be assessed through the signature block found in Annex E.

10. References. A8, A26.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Deliver a Tour and Operate the Anti-Fouling System					
1.1 Operate Anti-Fouling System to include: a. Identify location of Anti-Fouling unit and power supplies; b. Adjust output current for alongside function; c. Adjust output current for at sea function; and d. Check for system operation.	D	AP	P	A8	
TP2 Operate the Degaussing System					
2.1 Setting of Degaussing system	D	AP	P	A26	

EO 002.04 Inspect Motors & Controls

1. Performance. Inspect Motors & Controls.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log books.

   b. Denied: Nil.

   c. Environmental: Ship alongside or at sea.

3. Standard. The trainee shall Inspect Motors & Controls, IAW refs, by:

   a. Inspecting, electric motor (AUT1690); and

   b. Flashing up motor controllers locally and remotely.

4. Time Required.

   a. Completed during OJPR.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Self-study; and

   c. OJT.

6. Training Aids.

   a. Operational MCDV.

7. Training Limits. Ship Availability.

8. Task Numbers. AUT1690.

9. Test Details.

   a. There is no formal EC for EO 002.04. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A20.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Inspect Operating Running Machinery					
1.1 Verify running machinery, by: a. Checking for abnormal temperature b. Checking for abnormal vibration c. Checking for abnormal sound	D	AP	P	A20	Homework assignment to include all TPs.
TP2 Deliver Tour of Motor Controllers					
2.1 Describe motor controllers to include: a. Identifying system operation b. Start permissives c. Trips of; (1) Z-Drive (2) SCR Cabinet (3) Main fire pumps (4) Lube oil pumps (5) Fuel pumps (6) Compressed Air System (7) Blackwater (8) HVAC d. Carrying out lamp test	D	AP	P	A20	

TP3 Flash up Motor Controllers					
3.1 Flash up motor controllers from all locations to include: a. Z-Drive b. SCR Cabinet c. Main fire pumps d. lube oil pumps e. Fuel pumps f. Compressed Air System g. Blackwater h. HVAC	D	AP	P	A20	

PERFORMANCE CHECK – PO 002

1. Trainee will be provided feedback through PO 002. To achieve the intended assessment of PC 002, the trainee must achieve a qualifying signature in every signature block associated to PO 002 in the OJT package found in Annex E.

PO 003 Perform DC System Rounds

EO 003.01 Inspect DC Systems

1. Performance. Inspect DC Systems.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision

   b. Denied: Nil.

   c. Environmental: Ship alongside or at sea.

3. Standard. The trainee shall Inspect DC Systems, IAW refs, by:

   a. Configuring fitted firefighting controls (IT0620);

   b. Inspecting visually, smoke evacuation system components (AOT0670) by:

      (1) Testing smoke detector (ANT1430); and
      (2) Operating Main Machinery Space Camera's (HT0010).

   c. Configuring bilge eductor locally (ET0140).

4. Time Required.

   a. Completed during OJPR.

5. Instructional Methods.

   a. Demonstration – performance; and

   b. Self-study.

6. Training Aids.

   a. Operational MCDV.

7. Training Limits. Ship Availability.

8. Task Numbers. ANT1430, AOT0670, ET0140, HT0010, IT0620.

9. Test Details.

   a. There is no formal EC for EO 003.01. The content within this EO will be assessed through the signature block found in Annex E.

10. References. A11, A20, A31, A32, A33, A34, A35, A36, A37, C24.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Deliver a Tour of the Fire Fighting Control Systems					
1.1 Demonstrate all modes of activation for the: a. AFFF system; b. Karbaloy system; c. Flood and Spray system; d. CO2 system; and e. NOVEC system.	D	AP	P	A20, A31, A32, A33, A34	Must be able to describe system components, operating parameters, activation methods, health considerations, and post activation procedures.
TP2 Deliver a Tour and Operate the Smoke Evacuation System					
2.1 Smoke evacuation system layout to include; a. Location of all manually operated dampers; and b. Location of all fans.	D	AP	P		

2.2 Activation methods to include: a. Location of local activation switches; and b. Remote activation.					
2.3 Portable ventilation fans to include; a. Location of all portable fans; c. Location of all temporary trunking; and d. Layout trunking.					
2.4 Testing detection devices to include: a. Smoke detectors; b. Isolating Detectors and Zones; c. Locate; and d. Operate Main Machinery Space camera system	D	AP	P		
TP3 Deliver a Tour and Operate the Bilge and Ballast System Eductor System					
3.1 Bilge and Ballast System components to include: a. Manifolds b. Space suction valves c. Bilge/Ballast pump d. Eductor e. Rope stores hand pump f. Emergency suction from DA's g. Direct suction GSP/H&F pump h. Dirty oil Pump i. Ballast tanks	D	A P	P	A35, A37, A20, C24, A36	Qualified SS to walk through Bilge and Ballast system identifying key components.

EO 003.02 Inspect Fire main System

1. Performance. Inspect Fire main System.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References; and
      (6) Supervision

   b. Denied: Nil.

   c. Environmental: ship alongside / at sea.

3. Standard. The trainee shall Inspect Fire main System (ET0080) IAW refs, by:

   a. Configuring seawater service system (fire main) locally (ET0210).

4. Time Required.

   a. Completed during OJPR.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Guided discussion;

   c. Self-study; and

6. Training Aids.

   a. Virtual ship; and

   b. Operational MCDV.

7. Training Limits. Ship Availability.

8. Task Numbers. ET0080, ET0210.

9. Test Details.

   a. There is no formal EC for EO 003.02. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A11, A20, A38, C24.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Configure the Fire Main System					
1.1 Configuring fire pumps to include: a. Local/remote activation; b. Prestart and shut down procedures; and c. Running checks.	D	AP	P	A11, A20, C24	Qualified SS to walk through Bilge and Ballast system identifying key components.
1.2 Emergency Fire main procedures to include: a. Emergency bilge suction; b. split the firemain in half at fm40; c. Location and operation of the Tohatsu portable fire pump; d. Supply the firemain with the Tohatsu portable fire pump; and e. Tohatsu post operation maintenance.					
1.3 Testing/PM routines.					

PERFORMANCE CHECK – PO 003

1. Trainee will be provided feedback through PO 003. To achieve the intended assessment of PC 003, the trainee must achieve a qualifying signature in every signature block associated to PO 003 in the OJT package found in Annex E.

PO 004 Perform Propulsion System Rounds

EO 004.01 Inspect Fuel & Oil Systems

1. Performance. Inspect Fuel & Oil Systems.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log books.

   b. Denied: Nil.

   c. Environmental: ship alongside / at sea.

3. Standard. The trainee shall Inspect Fuel & Oil Systems IAW refs, by:

   a. Flashing up fuel oil system locally by (DT0140):

      (1) configuring centrifuge (GT0070); and
      (2) configuring the coalescer.

   b. Transferring main lube oil filling & transfer system locally (DT0160); and

   c. Configuring bilge pumping system (IT0550).

4. Time Required.

   a. Completed during OJPR

5. Instructional Methods:

   a. Demonstration – performance;

b. Self-study; and

c. OJT.

6. Training Aids.

a. Operational MCDV.

7. Training Limits. Ship Availability.

8. Task Numbers. DT0140, DT0160, GT0070, IT0550.

9. Test Details.

a. There is no formal EC for EO 004.01. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A20, A35, A39, A40, A41, A162.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Deliver a Tour and Operate the Fuel Oil Systems					
1.1 Location and operation of the F/O service components to include: a. Boost pumps; b. Return lines and valves; c. Service tank return/supply valves; and d. Bypass valves.	D	AP	P	A39	Qualified SS to walk through trainees on the system. Trainees to operate boost pumps and switch over service tanks.
1.2 Location and operation of the F/O fill and transfer components to include: a. Quick closing valves; b. Tank side valves; c. Hand pump; d. Transfer pump; e. Coalescers; f. F/O manifold; and g. Fuelling station.	D	AP	P	A39	Qualified SS to walk through trainees on the system. Trainees to operate transfer pump, hand pump, coalescers, and purifier.

TP2 Deliver a Tour of the Lube Oil Storage System					
2.1 Main Lube Oil Fill and Storage System components to include: a. Valves b. Storage tank; and c. Portable lube oil transfer pump.	D	AP	P	A39, A40	Qualified SS to walk through system and demonstrate procedure to fill tank.

EO 004.02 Inspect Shaft & Gearing

1. Performance. Inspect Shaft & Gearing.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log books.

   b. Denied: Nil.

   c. Environmental: ship alongside at sea.

3. Standard. The trainee shall Inspect Shaft & Gearing IAW refs, by:

   a. Verifying steering operations in follow up and non-follow up mode;

   b. Flashing up z-drive hydraulic oil (steering) system locally (XT0100);

   c. Flashing up Z-drive lube oil system locally (ABT0240);

   d. Operating main shaft line brake in local (ABT0180); and

   e. Configuring main seawater circulation system locally (DT0190).

4. Time Required

   a. Completed during OJPR.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Self-study; and

c. OJT.

6. Training Aids

a. Operational MCDV.

7. Training Limitations. Ship Availability.

8. Task Numbers. ABT0180, ABT0240, DT0190, XT0100.

9. Test Details

a. There is no formal EC for EO 004.02. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A20, A71, A94.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Describe Z-Drive Hydraulic and Lube Oil Systems					
1.1 Describe purpose, main components, layout, operating parameters, and modes of operation of: a. Z-Drive steering system; and b. Z-Drive Lube Oil system.				A20, A71, A94	Homework assignment.
TP2 Describe the Main Seawater Circulation System					
2.1 Describe purpose, main components, layout, and operating parameters of the Main Seawater Circulation System.				A20	
TP3 Deliver a Tour and Flash up Z-Drive Systems					

3.1 Lube oil system a. Main lube oil pump b. Standby lube oil pump 3.2 Hydraulic oil system a. Main hydraulic steering pump b. Standby hydraulic steering pump	D	AP	P	A20, A71, A94	Qualified SS to walk through Z-drive lube oil and hydraulic oil system and demonstrate.
TP4 Verify steering system operation					
4.1 Verify steering system operation at all positions a. Bridge; b. Switchboard room; c. Z-drive compartment	D	AP	P	A20	Qualified SS to demonstrate modes of operation for steering.
TP5 Operate the Shaft Brake					
5.1 Operate shaft brake locally	D	AP	P	A20	Qualified SS to show location of brake and methods to operate it.
TP6 Deliver a Tour and Flash up the Main Seawater Circulation System					
6.1 Main Seawater Circulation components to include: a. Sea bays b. Valves and cross connects c. Main DA d. Aux DA e. SCR heat exchanger f. Main motors g. Z-drive hydraulic and lube oil coolers h. Thyristor cooling pumps i. Thyristor cooling bypass switch	D	AP	P	A20	Qualified SS to walk through system and locate all valves and components in the system. Trainees to test Thyristor Pump standby function and bypass.

EO 004.03 Inspect Combustion Engines

1. Performance. Inspect Combustion Engines.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log books.

   b. Denied: Nil.

   c. Environmental: ship alongside / at sea.

3. Standard. The trainee shall Inspect, diesel engine by (QT0160), IAW refs, by:

   a. Flashing up Auxiliary Alternator;

   b. Flashing up Main Alternator;

   c. Flashing up Emergency Alternator; and

   d. Inspecting dirty oil system.

4. Time Required.

   a. Completed during OJPR.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Self-study; and

   c. OJT.

6. Training Aids.

   a. Operational MCDV.

7. Training Limits. Ship Availability.

8. Task Numbers. QT0160.

9. Test Details.

   a. There is no formal EC for EO 004.03. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A20, A41, A95, A96.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Flash up the Propulsion and Auxiliary Diesel Engines					
1.1 Flash up the Diesel Engines by conducting pre-start checks and undergoing flash-up and shut-down procedures. 1. 2 Auxiliary Diesel Alternator a. Pre-start checks b. Operating parameters c. Alarms d. Trips e. Flash-up and shut-down procedures 1.2 Emergency Diesel Alternator a. Pre-start checks b. Operating parameters c. Alarms d. Trips e. Flash-up and shut-down procedures 1. 3 Main Diesel Alternator a. Pre-start checks b. Operating parameters c. Alarms d. Trips e. Flash-up and shut-down procedures	D	AP	P	A20, A95, A96	

TP2 Deliver a Tour of the Dirty Oil System					
2.1 Locate components of the Fuel stripping system	D	AP	P		Qualified SS to walk through dirty oil system and indicate procedure for fuel stripping.
2. 2 Locate components of the Oily Water Separator System					

PERFORMANCE CHECK – PO 004

1. Trainee will be provided feedback through PO 004. To achieve the intended assessment of PC 004, the trainee must achieve a qualifying signature in every signature block associated to PO 004 in the OJT package found in Annex E.

PO 005 Respond to Emergency Conditions

EO 005.01 Respond to Emergency Conditions

1. Performance. Perform At-Sea Emergency Procedures.

2. Conditions.

   a. Given:

      (1) System schematics;
      (2) Access to systems;
      (3) Required tools;
      (4) PPE;
      (5) References;
      (6) Supervision; and
      (7) Applicable log book.

   b. Denied: Nil.

   c. Environmental: ship alongside / at sea.

3. Standard. The trainee shall Perform At-Sea Emergency Procedures, IAW refs, by:

   a. Conducting emergency engineering drills;

   b. Respond to Action stations; and

   c. Respond Emergency stations.

4. Time Required.

   a. Completed during OJPR.

5. Instructional Methods.

   a. Demonstration – performance;

   b. Self-study;

   c. OJT.

6. Training Aids.

   a. Operational MCDV.

7. Training Limitations. Ship Availability.

8. Task Numbers. Nil.

9. Test Details.

   a. There is no formal EC for EO 005.02. The content of this EO will be assessed through the signature block found in Annex E.

10. References. A37, A60, A95.

11. Remarks. Nil.

12. Teaching Points.

Teaching Point	Delivery Approach (Steps)			Refs	Comments Supporting the Instructional Strategy
	1	2	3		
TP1 Roundsperson Duties During Engineering Difficulties and Emergencies					
1.1 Drills a. Z-drive lube oil failure; b. Z-drive hydraulic oil failure; c. High Bearing temperatures; d. Motor enclosure fires; e. Steering gear failure; f. Fuel boost pump (one or both) failure; g. Blackout; (i) 600vAC (ii) 440vAC h. Main DA Drills; i. Loss of HP air; j. Loss of SIC k. Loss of thyristor cooling pump(one or both) 1.2 Action Stations a. Closing down to Zulu 1.3 Emergency Stations a. Fire; and b. Flood.	D	AP	P	A37, A60, A95	Qualified SS to demonstrate responsibilities of a roundsperson during drills, emergencies and conduct practical exercises.

PERFORMANCE CHECK – PO 005

1. Trainee will be provided feedback through PO 005. To achieve the intended assessment of PC 005, the trainee must achieve a qualifying signature in every signature block associated to PO 005 in the OJT package found in Annex E.

ANNEX A – MAIN REFERENCES

A = CF or other official Canadian government publication

B = Official Allied Forces or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	C-93-010-007/NY-001 *Batteries, Lead Acid, Vented*
A2	C-93-010-007/NY-Z01 *Batteries, Lead Acid, Vented*
A3	*All Class Ships SEMS*
A4	*All Class Ships MARCORD 43-02*
A5	C-27-877-000/MS-001 *Battery Charger for DDFP*
A6	CF-670 *Log for Secondary Battery*
A7	C-38-104-000/MS-001 *NATO Degaussing Folder*
A8	C-38-138-000/MS-001 *Cathodic Protection Manual*
A9	D-03-003-005/SF-001 *General Electrical Specification for Canadian Forces Ships*
A10	C-93-011-006/PT-001 *Americans Electricians handbook*
A11	C-23-MCD-000/MB-001 TYPE II OPERATIONAL STATIONS BOOK
A12	***Kingston Class*** CAMS Operator User Manual and Training Guide
A13	***Kingston Class*** Liquid Cargo Report
A14	C-27-B20-000/MB-001 Freshwater System
A15	C-26-325-000/MS-001 *Casualty Power System Components*
A16	C-27-B21-000/MF-001 Operation and Maintenance Manual for Zenon Reverse Osmosis Desalinator Unit SCW 330C
A17	C-27-B53-000/MB-001 Black Water System
A18	C-27-B56-000/MB-001 Grey Water System
A19	C-27-B26-000/MB-001 Compressed Air System
A20	Kingston Class Engineering Officers Technical Instructions (EOTIs).
A21	C-97-380-AA0/MS-001 OPERATIONS MANUAL - HIGH PRESSURE BREATHING AIR COMPRESSOR UNIT C-D/DV/NAVY/SS (APPLICABLE TO HFX, IRO, PTR, KIN, CFNES AND CFFS(E))
A22	C-29-503-000/MB001 MCDV Food Refrigeration System operational and Maintenance Manual
A23	C-29-510-000/MF001 Air Handling Unit Installation and Maintenance Manual
A24	C-29-511-A00/MF001 Marine Air Conditioning Unit Instructional Manual
A25	C29-502-000/MF-001 MCDV Condensing Unit
A26	C-38-232-000/MS-002 *Degaussing Power Supplies*
A27	102V3H Maintenance Plan Narrative System Description
A28	C-26-484-000/MF-001 Main Power and Propulsion, Auxiliary and Emergency Switchboards, Console and PGSCS Manual
A29	C-51-714-000/MF-001 Whistle System
A30	C-57-742-000/MF-001 N-D2000 Navigation Light Control Panel.

A31	C-97-337-000/MB-001 Aqueous Film Forming Foam System
A32	C-97-340-000/MB-001 MCDV Galley Hood Fire Suppression System
A33	C-97-339-000/MB-001 Magazine Sprinkler System
A34	C-77-344-000/MB-001 Propulsion Motors CO2 System
A35	C-27-B19-000/MB-001 Bilge and Ballast System
A36	C-77-149-000/MF-001 A.C. Portable Submersible Pump
A37	CFCD 132 Damage Control Manual Vol 2
A38	C-27-B17-000/MB-001 Saltwater System
A39	102V5G MCDV Fuel Oil System
A40	C27-B22-A00/MF-001 Quick Closing Valve System
A41	C-27-B57-000/MB-001 Lubricating Oil System
A42	C-23-343-000/MS-001
A43	C-23-343-000/NY-002
A44	C-23-343-000/NY-Z05
A45	C-03-010-546/MS-001 *AC motor controllers*
A46	C-23-050-001/PT-001 *Lab Volt Student Manual*
A47	C-23-050-001/PT-001 *Lab Volt Student Manual*
A48	D-01-003-001-S6-000 *Drawing standards*
A49	D-01-400-001-S6-000 *Drawing standards*
A50	C-38-288-000/MS-001 *Frequency Converter*
A51	C-28-463-000/MS-001 *Sliding Padeye*
A52	C-23-343-000/MS-001 *RHIB*
A53	C-23-045-000/AG-001 *Damage Control Manual Vol. 1*
A54	**None Allocated**
A55	C-03-005-012/AM-001 *NaMMS Vol. 1*
A56	**None Allocated**
A57	C-03-005-012/AM-002 *NaMMS Vol. 2*
A58	*Preventive Maintenance Control Documentation*
A59	**None Allocated**
A60	*Maritime Command Ships Standing Orders*
A61	*MARCORDS*
A62	C-28-010-329/MS-001 *Watertight Closures*
A63	C-03-011-007/MS-001 *Insulation Manual*
A64	**None Allocated**
A65	D-23-003-005/SF-002 *Paint Specifications for HMC Ships*
A66	**None Allocated**
A67	D-28-155-000/SF-003 *Deck Covering Material Specifications*
A68	**None Allocated**
A69	D-49-003-003/SF-001 *Welding Specifications for HMC Ships*

A70	D-03 001-025/SF-001 *Specifications for Repair and Overhaul Marine Valves*
A71	C-24-656-000/MF-001 LIPS Steerable Thruster Manual Vol. 1
A72	C-03-005-033/AA-000 *Engineering Manual Vol. 1*
A73	D-23-003-005/SF-001 *Specification for the Preservation and Painting of HMC Ships (New Construction)*
A74	C-28-177-000/TB-004 *Technical Bulletin (Marine) Bushing Arrangement for Dog Assembly for W/T Individual Dogged Door*
A75	C-28-177-000/TB-003 *Bushing Arrangement Dog Assembly*
A76	C-03-010-104/ME-000 *Maintenance of Valves and Fittings*
A77	C-25-007-001/TB-001 *Velan Steam Traps*
A78	**None Allocated**
A79	C-28-177-000/TB-001 *Hinge Pin Assembly – Teflon Bushing Heintz Door*
A80	**None Allocated**
A81	C-28-177-000/TB-002 *Lever Hold Back Clip Watertight Quick-acting Door*
A82	**None Allocated**
A83	C-97-007-001/TB-001 *Twin Agent Unit*
A84	**None Allocated**
A85	C-97-040-001/TB-001 *Dry Chemical Extinguisher*
A86	**None Allocated**
A87	A-P9-000 SERIES *PERS Occupation Specifications*
A88	C-29-104-000/MS-001 *Instruction Book and Parts Manual for Air Conditioning*
A89	C-77-152-000/MS-001 *NBC Filtration Units 280 Class*
A90	C-29-131-000/MJ-001 *Air Conditioning 280's*
A91	C-28-010-024/MS-001 *Anchors, Cables and Associated Fittings*
A92	**None Allocated**
A93	C-29-050-004/PT-001 *Marine HVAC Theory*
A94	C-24-656-000/MF-002 LIPS Steerable Thruster Manual Vol. 2
A95	***Kingston Class*** Emergency Operating Procedures
A96	***Kingston Class*** Technical Instructions
A97	**None Allocated**
A98	
A99	
A100	C-29-131-000/PT-000 *HVAC Training Manual*
A101	**None Allocated**
A102	C-LM-181-001/JS-001 *CFP181*
A103	**None Allocated**
A104	A-PD-152-001/PT-001 *Seaman's Handbook*
A105	**None Allocated**

A106	C-23-045-000/AG-003 *Damage Control Manual Vol. 3*
A107	D-03-003-016/SF-005 *Compartment Testing*
A108	**None Allocated**
A109	C-28-020-001/TB-001 *Shipboard Lifting Appliance*
A110	**None Allocated**
A111	C-34-010-002/AM-000 *Rigging Manual*
A112	**None Allocated**
A113	D-03-010-001/SF-001 *Furniture and Finishing Material for HMC Ships*
A114	**None Allocated**
A115	C-27-010-016/TB-001 *Decontamination of Fresh Water Tanks*
A116	**None Allocated**
A117	**None Allocated**
A118	
A119	D-03-002-004/SG-007 *Ships Standard – Non-Standard Pipe Work Components*
A120	**None Allocated**
A121	**None Allocated**
A122	*Captain's Ships Book*
A123	C-38-138-000/MS-001 *Cathodic Protection Manual*
A124	**None Allocated**
A125	CFAO 30-2 *Fire Reporting and Investigation*
A126	**None Allocated**
A127	*QR&O's*
A128	**None Allocated**
A129	C-23-045-000/AG-002 *Damage Control Manual Vol. 2*
A130	**None Allocated**
A131	C-02-015-001/AG-000 *Unsatisfactory Condition Report – CF 777*
A132	**None Allocated**
A133	C-03-005-034/TB-001 *Preventive Maintenance Amendment Proposal – CF 507*
A134	**None Allocated**
A135	A-PD-121-C01/FP-000 *DND Administrative and Staff Procedures Manual*
A136	**None Allocated**
A137	A-AE-000-001/AG-E00 *Guide to the Divisional System*
A138	**None Allocated**
A139	C-03-005-012/AM-007 *NaMMS Vol 7*

A140	**None Allocated**
A141	C-03-040-000/TS-001 *MARCOM Precautions and Safety Precautions for Welding and Burning*
A142	**None Allocated**
A143	**None Allocated**
A144	C-94-010-03/MG-000 *Compressed Gas Cylinders*
A145	**None Allocated**
A146	**None Allocated**
A147	
A148	C-77-010-008/MB-001 *Incident and Status Boards for HMC Ships*
A149	**None Allocated**
A150	D-23-002-005/SG-001 *Standard for Metric Draft Marks*
A151	**None Allocated**
A152	C-28-199-000/MJ-000 *Instruction Book and Installation Manual for Hangar Doors*
A153	**None Allocated**
A154	***Orca Class*** *Technical Package*
A155	***Orca Class*** *Standing Orders*
A156	***Orca Class*** *PCTU General Bridge Pack*
A157	***Orca Class*** *Know Your Ship Book*
A158	***Kingston Class*** *Manufacturer's Operating Manuals*
A159	***Kingston Class*** *Isherwood Software*
A160	***Kingston Class*** *Training Data Package*
A161	***Kingston Class*** *MDTU Operator's Manual*
A162	***Kingston Class*** *SEMS*
A163	***Kingston Class*** *Know Your Ship Book*
A164	***Kingston Class*** *Engineering Logs*
B1	**None Allocated**
C1	ISBN 9780827341463 *Direct Current Fundamentals 4th edition*
C2	ISBN 1418065803 *Delmars Standard Textbook of Electricity 4th edition*
C3	ISBN 0321374177 *Introduction to Technical Mathematics 5th edition*
C4	*Alternating Current Fundamentals 4th edition*
C5	*Individuals Equipment Operator's Manual*
C7	ISBN 0-02-342-700-0 *Technical Drawing (Sixth Edition)*
C8	ISBN 0-87006-577-7 *Modern Woodworking*
C9	ISBN 0-13-345471-1 *Furniture and Cabinet Construction*
C10	ISBN 87345-104-X *Metalwork Technology and Practice*
C11	ISBN 0-87006-429-0 *Blueprint Reading*
C12	ISBN 0-87006-279-4 *Modern Welding*

C13	ISBN 0-07-548922-8 *Engineering Drawing and Design (Third Edition)*
C14	**None Allocated**
C15	**None Allocated**
C16	ISBN 0-13-898762-2 *Technically Write*
C17	ISBN 0-87006-509-2 *Modern Metalworking*
C18	ISBN 0-07-082462-2 *General Metals*
C19	*Range Guard Installation and Maintenance Manual*
C20	ISBN 87002-307-1 *Wood Materials and Processes*
C21	ISBN 0-471-80043-0 *Mathematics for the Trades*
C22	*Hilti Gun Manufacturers Instructions – DX-600N*
C23	*Arcair Slice Cutting System Instruction Manual Thermadyne Arcair Company*
C24	TOHATSU VC72AS Fire Pump Portable Manual
D1	*Mar Tech JTAR #600404*

ANNEX B – TASK LIST

Task No	Task Statement	Trg/No Trg
See JTAR and individual Chapter 2 POs for task numbers	The JTAR board designated tasks from the RQ-S3 task list, these were deemed class specific and duty watch tasks. These tasks were used for the base to build the roundsperson POs. Other RQ-S3 tasks were imported from the task list in order to fully capture a roundspersons' duties. While the core knowledge was trained on the RQ-S3, the class specific aspects of these tasks were used to build class packages.	Decisions are captured as Comments in the JTAR where an NT occurred. NT tasks were usually decided on redundancy or they were covered at another level such as NETP.

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 09 Jan 2017 – 03 Feb 2017

Location: WP1020 (Haida –Venture)

Board Members:

Surname	Initials	Rank	MOSID	Unit
Wilson (Chair)	R.P.	Lt(N)	00345	NFS(P)
Boyd (TDO)	R.	Civilian		NTDC(P)
Moazzezi Moghaddam (OPI)	A.	Lt(N)	00345	NTDC(P)
Gagne (NPTG Rep)	Y.S.	CPO2	00124	NPTG HQ
Brake (SME)	W.F.	Lt(N)	00345	NTDC(A)
Ahn (Secretary)	H.J.	A/SLt	00345	NFS(P)
Tall (Secretary)	T.L.	LS	00124	NFS(P)
Hockey (SME)	M.L.	CPO2	00225	F4 ENG
Moody (SME)	M.E.	CPO2	00125	NTDC(P)
Lambert (SME)	S.J.	CPO2	00367	NTDC(P)
Robinson (SME)	M.J.	CPO2	00124	NTDC(P)
Penney (SME)	K.G.	PO1	00125	NTDC(P)
Lester (SME)	D.I.	PO1	00367	NTDC(P)
Medal (SME)	R.W.	MS	00124	NTDC(P)
Garis (SME)	M.D.	MS	00367	FAP(P)
Brown (SME)	S.S.	LS	00125	HMCS FREDERICTON
Kale (SME)	C.	LS	00367	HMCS VANCOUVER
Thorsteinson (SME)	P.	LS	00124	NFS(P)

Board Objective:

Produce the Basic Maintainer and Roundsperson qualification. Produce the appropriate the Roundsperson Ship Class qualification. Identify the harbour MCC/DCC package requirements.

Specification Task – No Train Decisions:

Decisions are captured as Comments in the JTAR where an NT occurred. NT tasks were usually decided on redundancy or they were covered at another level such as NETP.

Discussions/Decisions:

14 Feb 17 – While the OS RQ QSP was being formatted, the board continued on to producing the QSPs for the class packages. During the formatting process, this QSP was reviewed many times, and minor changes were made to optimize training. The board acknowledges that all the annexes are not complete. For example, references will be fleshed out during the course content development phase. The costing estimate was reviewed on several occasions, and estimates are as close as the board can make. On this day, the board formally reviewed this QSP and agreed on the contents in the ROPs. This OS QSP is now complete, and the board will continue on with the class packages for the Halifax, Kingston, and Orca roundsperson packages.

06 Feb 17 – Upon completion of the OS-QSB the board members moved on to writing the QSPs for the Class Packages (CP). During the first two days, there was considerable discussion on how to organize the QSPs for the three different classes; Halifax, Kingston and Orca. Available QSPs and various existing roundsperson packages as well as IMO material was reviewed. There was considerable discussion around various points, such as whether to provide sketches or have the students sketch. The board considered many ways to shorten the roundsperson course and to have the majority of testing done under the direction of the school. It was decided that only the PCs for each PO would be conducted on the ship in order to reduce time spent on ship working on an OJT. After general discussions, the board commenced the task analysis (TA).

Having just completed the OS QS the board was very aware of the TA process. Scalars for all three CPs were constructed at the same time in order to ensure that tasks were considered for each CP. The SMEs worked the POs into a system-based approach, this allowed for easy placement of all tasks. During TA tasks were reorganized several times until the final product was decided upon. The board decided on five Marine System Engineering (MSE) oriented POs and one Combat System Engineering (CSE) oriented PO. PO006 (CSE PO) was derived from the MCR Watch Package. The MCR Watch Package was reviewed to find which tasks were not already covered in the MSE based POs. The board decided on three EOs from the MCR watch Package to be trained. Subsequent discussions with CSE SME reduced that number to two items. Lesson plans for these items are planned to be drawn from existing CSE materiel and inserted into the CP instruction where it fits best to accommodate scheduling.

The board broke into groups in order to build the EOs and teaching points for their respective occupation.

16 Feb 17 – The board came together to collect all EOs to be entered into the master documents by the board secretary. There was discussion on how the OJPR for testing of the PCs was to be constructed. As the school is already testing ECs only a signature for each PO will be required onboard ship. The OJPR signature sheet will be included as an Annex in each QS. Work continued on making the EOs and Teaching Points.

23 Feb 17 – Work continued on entering all the requisite information into the class package documents. The CP information at this time was finalized and the CPs themselves complete with respect to content, only formatting and completion of necessary annexes remained.

Lessons Learned:

See above comments

Guidance to Instructional Staff:

See applicable EOs in respective POs.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1			
2			

Other Considerations (if applicable):

ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

_____

Name and Initials Rank SN

_____ Date

Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. Recording of the individual's progress shall be entered in Monitor Mass under the Training Tracker. The NFS for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance Objective to the required standard, the MSE Training PO will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and enter progress in the Monitor Mass Training Tracker.

3. If it is not possible to conduct OJPR for some POs due to equipment not fitted or not issued (i.e. Work Period), the MSE Training PO is directed to contact NFS (P) for guidance. Extension of training time might be required.

4. See Appendix 1 for OJPR sign off list.

5. Appendix 1's Performance Checks require the trainee to demonstrate the listed competencies as described in chapter 4. A series of questions will be asked from each EO to satisfy the knowledge of the rounds person before granting qualification. The trainee will have three (3) months from being posted to an operational ship to complete Appendix 1.

6. On completion of this OJPR, the MSE Training PO shall email (scanned) the *Certificate of Qualification* for MITE update to;

   a. +NFS(A) MITE@NFS(A)@Halifax; or

   b. +NFS(P) MITE@NFS(P)@Esquimalt.

7. If you wish to make a change, send your request to: +NPTG MITE@NPTG HQ@Esquimalt

   PROGESS REVIEW

8. Progress review shall be carried out quarterly by the Divisional Officer and Ch Eng.

OJPR Reviewed By (Rank, Position, Name)	Signature	Date

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

Chief Engineer/ MSEO/ CSEO (Print and sign) Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in the Roundsperson QSP-OJPR for the KIN class vessel Kingston Class Package, (except as noted). The member is qualified to carry out the duties of a Roundsperson on a KIN class vessel as a Roundsperson and is NQual ALMK qualified.

_____

Divisional Officer (Print and Sign) Date

ON-JOB PERFORMANCE RECORD SHEET

1. The remarks column in Appendix 1 shall be used for the following information only:

   a. The unit in which the training and assessment were carried out; and

   b. Any unusual circumstances which preclude compliance with an objective.

APPENDIX 1

PO/EO	Description		Date	Signature	Remarks
PO 001 Perform Auxiliary System Rounds					
EO 001.01	**Inspect Liquid Handling Systems**				
	Local level monitoring of sound tubes / level indicators / LSU /CAMs and their locations to include: a. Ballast; b. Potable water; c. Blackwater; d. Greywater; e. Void; f. DFO; g. Dirty Oil; h. Oily water; and i. Lube Oil storage.	TP2.1 – Tank level monitoring			
EO 001.02	**Inspect Domestic Services**				
	a. Procedure for filling the FW system; b. Procedure for changing over tanks; c. Procedure for priming and flashing pumps. d. Local start/stop of the calorifiers.	TP3.2 – Operate the Freshwater services system			
	a. Configure the system to treat either tank; b. Procedure for treating a FW Tank; c. Adjusting and testing bromine levels.	TP3.3 – Operate the Bromine purification system			
	a. Changing Cartridge filters, clean strainers; b. Alarm points and trips; c. Conducting rounds; d. Flash-up and shutdown; and e. Freshwater flush and cleaning.	TP3.4 – Operate the Shipboard Reverse Osmosis Desalination (SROD)			

	a. Treated black water system operation, collection / overboard (Chlorine and UV); b. Procedure for discharging ashore/at sea.	TP4.3 – Operate the Black water system			
	a. Configuring the grey water system between inboard (Collection) and outboard (Discharge) use.	TP4.5 – Operate the Grey water system			
EO 001.03	**Inspect Compressed Air System**				
	a. Flash-up of air compressor; b. Alarms and trips; c. Lead / Lag set points (switching over lead and lag compressors); d. Draining Flasks; e. Rounds on the Compressor; and f. Reducing stations.	TP2.2 – Operate the HP air system			
	a. Fill bottles from the Bauer portable compressor.	TP3.1 – Operate Breathing Air compressor			
EO 001.04	**Inspect Refrigeration & A/C**				
	a. Locate Parasense location(s); b. Flash up refrigeration system; c. Identify Alarms and trips; and	TP2.2 – Operate the Main Refrigeration System			
	a. Procedure for local/remote operation of A/C Units; b. Changing filters on units; and c. Procedure for changing between Heat / AC / Recirculation.	TP3.1 – Operate the HVAC system			

PO 002 Perform Electrical System Rounds					
EC 002.01	**Inspect PG & D (Must be Assessed by Senior Electrician or Legacy QL5 Electrican)**				
	a. Shore power to ships power (MA and AUX DA); b. Ships power to shore power (MA and AUX DA); and c. Under direct supervision of the Snr. Electrician, Connect/disconnect shore power cables.	TP3.1 – Synchronize DA and Shore Power (Transferring ships load)			
	a. Receive power to a dead bus; and b. Place on load (Synchronize) two MDA's via the MCR console.	TP3.1 Switchboard Operations			
	a. Receive power to a dead bus; b. Synchronize Emerg DA and ships service switchboard (EBT1 and EBT2); and c. Restore/Reset switchboard and Emerg. DA after blackout operations.	TP3.4 – Emergency Switchboard Operations			
	a. Reset and restore after system activation.	TP3.2 Anti-blackout system			
	a. Describing an earth and its effects on personnel and equipment; b. Proper operation of the ground detection system; and c. Proper tracing and isolation of grounds.	TP3.5 – Ground fault isolation			
	a. Isolate power in response to emergencies in home port b. Foreign port.	TP3.6 Isolation of electrical power			

EC 002.02	**Inspect Electrical Systems**				
	a. Test At will switch locations; b. Timing circuits; and c. IFDMS integration switch to unmanned.	TP 1.2 – Operate the Fog horn			
	a. Test Man overboard alarm; b. Man locked in fridge; and c. IFDMS	TP 2.1 Verify shipboard alarms			
EC 002.03	**Inspect Cathodic & Degaussing System**				
	a. Switch between sea and harbor operation; and b. Monitor/rounds and record readings.	TP2.1 – Inspect/Operate the Cathelco anti-fouling system			
	a. Flash up degaussing system including the input settings as per NATO Folder.	TP3.1 – Operate Degaussing system			Not all MCDV's have units installed at the time of this OJPR.
PO 003 Perform DC System Rounds					
EC 003.01	**Inspect DC Systems**				
	a. Activating the system; b. Reset pull/valve; c. Hazmat consideration; and d. PM routine.	TP4.1 – Operate the AFFF system (Simulated, do not activate the AFFF system)			
	a. Remote activation; b. Local activation;	TP4.1 – Operate the Main Motor CO2 System (Simulated, do not activate the CO2 system)			
	a. Deployment and activation of the NOVEC system; and b. PM routine.	TP4.1 – Operate the NOVEC system			

		(Simulated, do not activate the NOVEC system)			
	a. Describe the method of automatic operation	TP4.1 – Operate the Range Guard (Karbaloy) system.			
	a. Describe the different methods of activation; and b. Describe the procedure for Testing/PM routine.	TP5.1 – Operate the Flood and Spray system. (Magazine spray, pyro and demo locker flood)			
	a. Shutdown ventilation from the MCR; b. Correctly confine smoke zone 1 / 2; c. Configure and flash up the RAMFAN;	TP2.3 – Operate the smoke evacuation system.			
	a. Describe what is required to obtain permission to utilize the bilge eductor system; and b. Describe how to configure the bilge eductor system to evacuate bilge contents	TP6.1 – Operate the bilge eductor system.			
EC 003.02	**Inspect Firemain System**				
	a. Local/remote activation for pumps; b. Running checks on pumps.	TP2.1 – Operate the Firemain System			
	a. Emergency bilge suction; b. Separate the firemain at fm40; and c. Rig and operate the Tohatsu portable fire pump to supply the firemain.				
PO 004 Perform Propulsion System Rounds					

EC 004.01	Inspect Fuel & Oil Systems				
	a. Manually open service tank suction and discharge valves; b. Configure fuel return valve to the correct service tank; c. Boost pump flash-up and shut down procedures; d. Configure boost pump into by-pass; e. Operate the quick closing valves remotely.	TP3.1 – Operate the Fuel Oil service system.			
	a. Flash up and shut down procedures of the fuel oil centrifuge; b. Flash up and shut down of the coalescer; c. Fuel Oil Transfer pump flash-up and shut down procedure; and d. Conduct rounds on FOT system.	T32.2 – Operate the Fuel Oil transfer system.			
	a. Lube Oil Transfer pump flash-up and shut down procedures; and b. Filling a diesel generator with oil.	TP4.1 – Operate the Diesel Lube Oil and Transfer System.			
	a. Set up the dirty oil pump to transfer oily water from one bilge to the oily water tank.	TP5.1 – Operate the Bilge System			
EC 004.02	Inspect Shaft & Gearing				
	a. Conduct a set of rounds on the Lube oil system; and Switchover filters.	TP3.1 – Lube Oil Z-Drive System			
	a. Conduct a set of rounds on the steering system.	TP3.2 – Operate the (Hydraulic) Steering Gear System			
	a. Complete steering checks at the primary; secondary steering; and Emergency steering positions.	TP 4.1 – Operate Steering System			

	Manual/locally operate the shaft holding brake.	TP5.1 – Shaft holding brake			
	a. Flash up the Saltwater (Thyristor) Cooling System; and b. Place the Thyristor cooling switch into by-pass	TP6.1 – Operate the Saltwater (Thyristor) Cooling System.			
EC 004.03	Inspect Combustion Engines				
	a. Conduct pre-start checks; b. Oil and coolant type; c. Flash up locally; d. Conduct a set of rounds; and e. Shutdown the engine, including resetting from an emergency shutdown procedure.	TP2.1 – Operate the Auxiliary (UD19) Diesel engine			After shutdown engine must be set for automatic/remote operation.
	a. Conduct pre-start checks; b. Oil and coolant type; c. Flash up locally (no load); d. Conduct a set of rounds; and e. Shutdown the engine, including resetting from an emergency shutdown procedure.	TP2.1 – Operate the Emergency (UD16) Diesel Engine			After shutdown engine must be set for automatic/remote operation.
	a. Conduct a set of pre-start checks; b. Oil and coolant type; c. Recharging freshwater coolant circuit, d. Flash up locally; e. Conduct a set of rounds; and f. Shutdown the engine, including resetting from an emergency shutdown procedure (barring over).	TP2.1 – Operate the Main (UD23) Diesel Engine			After shutdown engine must be set for automatic/remote operation.
PO 005 Respond to Emergency Conditions					

EC 005.01	Perform At-sea Emergency Procedures				
	Roundsperson duties during engineering emergencies	Z-Drive Lube Oil and Hydraulic oil failure(s)			
		High Bearing temperatures			Diesel engine and propulsion motor bearings.
		Propulsion Motor Enclosure fires			
		Respond to a DC Alarm (Fire/Flood)			Rapid Response Team (RRT)
		Steering gear failure to include: a. Changing over to secondary and emergency steering; and b. Reverting to primary steering.			
		Fuel system failure; a. Placing boost pump by-pass and returns to correct tank; and b. Conducting rounds of the system pressure.			
		Diesel Generator Drills			
		Respond to a DC Alarm (Fire/Flood)			
		Respond to a machinery space fire i.e. Isolate out the non-affected machinery space.			Conducted either as the longest off watch roundsperson or after RRT.
		Loss of HP air			
		PGSCS UPS Failures			

	Changing over power supplies	600vAC & 450vAC Failure(s)			Response to direction from Watchkeeper including resetting and re-flashing systems.
		SCR in local operation			
	Responding, acknowledging and resetting local alarm conditions	Steering gear alarms			
		Switchboard alarms			
		UPS alarms			
		SCR/shaftline failures			
		LSU alarms			

Name and Initials Rank Service Number

Training Started Date

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym and Abbreviation	Term or Definition
A/C	Air Conditioner
ACOS	Automatic Change Over Switch
AC	Alternate Current
AFFF	Aqueous
B/W	Black Water
CBRN	Chemical Biological Radiological Nuclear
D/C	Damage Control
DC	Direct Current
F/W	Fresh Water
FA	First Aid
FOS	Fuel Oil System
FOT	Fuel Oil Transfer
G/W	Grey Water
GT	Gas Turbine
HMI	Human Machine Interface
HP	High Pressure
HVAC	Heat and Ventilation System
LP	Low Pressure
MCOS	Manual Change Over Switch
MLO	Main Lube Oil
PDE	Pielstick Diesel Engine
PG&D	Power Generation & Distribution

PLC	Programmable Logic Controller
S/W	Sea Water
SFC	Static Frequency Converter
SSS	Synchronous Self Shifting
TAU	Twin Agent Unit
UPS	Uninterruptable Power Supply
W/T	Water Tight Door

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

Not Required. No residential instruction provided. See Chapter 3 and 4 for environment and timings.

ANNEX H – ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**)

No additional training resources identified at the end of the QS Writing Board.

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable (**Not to be deleted**).
Estimates only, based on Expenditure Summary by Activity and GL FY 2016/17 at NFS(P)

Course: RQ-OS

School Supplies/Textbooks	Unit of Issue	Unit Price	QTY	Total Cost
1" Binder	EA	$9.39	12	$112.68
12" Ruler	EA	$3.19	12	$38.28
Eraser	EA	$2.15	12	$25.80
Blue Pen	EA	$0.69	12	$8.28
Pencil	BOX	$6.39	1	$6.39
Lined Paper	PKG	$9.99	12	$119.88
Highlighter	BOX	$21.48	1	$21.48
Subtotal				**$332.79**
Hand Tools	Unit of issue	Unit Price	QTY	Total Cost
Miscellaneous replacement				$500.00
Subtotal				**$500.00**
Work & Protective Clothing	Unit of Issue	Unit Price	QTY	Total Cost
Safety Glasses	EA	$11.29	4	$45.16
Shop Clothing				$1000
Subtotal				**$1045.16**
Office Supplies (Qualified SSs)	Unit of Issue	Unit Price	QTY	Total Cost
Pens, Blue	EA	$2.69	12	$32.28
Pens, Red	EA	$2.69	12	$32.28
Pencils	BOX	$6.39	1	$6.39
Post it Notes	PKG	$5.89	4	$23.56
Writing Pad 8.5x11	EA	$4.79	4	$19.16
Whiteboard Marker	PKG	$6.99	1	$6.99

Whiteboard Eraser	EA	$11.99	1	$11.99
Paper Clips	BOX	$1.09	4	$4.36
Sharpie Marker	EA	$3.69	2	$7.38
Subtotal				**$144.39**
Total Per Course				**$2022.34**

ANNEX J – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details as to the nature and reasons for amendments are to be provided here.

Form found in NPTORD 4500-22 Annex B

NPT 500-22
Annex B_QSP Amendm

ANNEX K – ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria	
EC 001.01	EO 001.01	To assess the Marine Technician Basic Maintainers ability to Inspect Liquid Handling Systems	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 001.02	EO 001.02	To assess the Marine Technician Basic Maintainers ability to Inspect Domestic Services	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 001.03	EO 001.03	To assess the Marine Technician Basic Maintainers ability to Inspect Compressed Air Systems	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 001.04	EO 001.04	To assess the Marine Technician Basic Maintainers ability to Inspect Refrigeration and Air Conditioning	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54

PC 001	PO 001	To assess the Marine Technician Basic Maintainers ability to Perform Auxiliary System Rounds	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 002.01	EO 002.01	To assess the Marine Technician Basic Maintainers ability to Inspect Power Generation and Distribution.	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 002.02	EO 002.02	To assess the Marine Technician Basic Maintainers Inspect Electrical Systems	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 002.03	EO 002.03	To assess the Marine Technician Basic Maintainers ability to Inspect Cathodic and Degaussing Systems	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 002.04	EO 002.04	To assess the Marine Technician Basic Maintainers			TRB IAW

		ability to Inspect Motors and Controls	Signature Block IAW Annex E	OJT to be completed IAW Annex E	NPTORD 4500-54
PC 002	PO 002	To assess Marine Technician Basic Maintainers ability to Perform Electrical System Rounds	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 003.01	EO 003.01	To assess the Marine Technician Basic Maintainers ability to Inspect Damage Control Systems	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 003.02	EC 003.02	To assess the Marine Technician Basic Maintainers ability to Inspect Fireman System	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
PC 003	PO 003	To assess the Marine Technician Basic Maintainers ability to Perform Damage Control System Rounds	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 004.01	EO 004.01	To assess the Marine Technician Basic Maintainers	Signature Block IAW Annex E		TRB IAW

		ability to Inspect Fuel and Oil Systems		OJT to be completed IAW Annex E	NPTORD 4500-54
EC 004.2	EO 004.02	To assess the Marine Technician Basic Maintainers ability to Inspect Shaft and Gearing	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 004.03	EO 004.03	To assess the Marine Technician Basic Maintainers ability to Inspect Combination Engines	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
PC 004	PO 004	To assess the Marine Technician Basic Maintainers ability to Perform Propulsion System Rounds	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54
EC 005.01	EO 005.01	To assess Marine Technician Basic Maintainers ability to Perform At Sea Emergency Procedures	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54

PC 005	PO 005	To assess the Marine Technician Basic Maintainers ability to Respond To Emergency Conditions	Signature Block IAW Annex E	OJT to be completed IAW Annex E	TRB IAW NPTORD 4500-54

ANNEX L – TEL PLAN

PO	TEL Requirement	Cost	Impact on Training
PO 001	Compressed air system mock-up Various domestic system mock-ups	TBD	Reduce ship-board training
PO 002	Switchboard and generator mock-up Cathodic and Degaussing system mock-up	TBD	Reduce ship-board training
PO 003	N/A at this time	TBD	N/A
PO 004	Fuel transfer system mock-up Gearing mock-up	TBD	Reduce ship-board training
PO 005	N/A at this time	TBD	N/A